

# **Executive Business Report**

## **Retail Sales Forecasting & Inventory Planning**

### **Executive Summary**

This report analyzes historical retail sales to identify business trends, forecast future demand, detect unusual sales patterns, and recommend inventory strategies. After comparing three forecasting approaches, the XGBoost model delivered the most reliable predictions with an estimated forecasting accuracy of 82.96%, making it the preferred model for future planning. The analysis shows that Office Supplies will continue to generate the highest demand, while the West region is expected to outperform the East region. Eleven unusual sales periods were identified, highlighting the importance of monitoring exceptional business events such as seasonal promotions and supply shortages. Overall, the findings provide a strong foundation for improving inventory management, reducing stock-related risks, and supporting better business decisions.

### **Key Business Findings**

Sales show a steady long-term increase with recurring seasonal demand. XGBoost achieved the best performance (MAPE 17.04%, MAE 16,989.80, RMSE 19,665.69) compared with SARIMA and Prophet and is recommended for future forecasting.

### **Three-Month Sales Forecast**

Overall sales are expected to remain stable with gradual growth over the next three months. Normal month-to-month variations are expected, but the long-term trend remains positive. Forecast confidence ranges indicate that actual sales may vary moderately around the predicted values, so inventory should be planned with some flexibility.

### **Top Business Anomalies**

Isolation Forest detected 11 unusual sales weeks. High sales were likely caused by festivals, promotions, or discounts, while low sales may be linked to stock shortages, lower demand, or off-season periods. These events should be reviewed separately because they can influence forecast accuracy.

### **Product Demand Segmentation & Stocking Strategy**

Office Supplies: highest demand—maintain higher inventory and frequent replenishment. Furniture: stable growth—maintain balanced stock levels. Technology: slight decline expected—avoid overstocking and replenish based on demand. The West region is expected to outperform the East region; therefore, allocate additional inventory to the West.

## **Business Recommendations**

1. Prioritize Office Supplies inventory to reduce stockouts.
2. Allocate more inventory to the West region to match expected demand.
3. Use XGBoost as the primary forecasting model because it achieved the highest accuracy (82.96%).

## **Business Risk**

Unexpected events such as economic changes, supply chain disruptions, promotions, or changing customer behavior can reduce forecast accuracy. Forecasts should be reviewed and updated regularly.

## **Conclusion**

The analysis provides valuable insights into future sales, product demand, regional performance, and unusual sales behavior. Implementing XGBoost-based forecasting together with targeted inventory planning will help improve stock availability, reduce operational costs, and support sustainable business growth.